

Network Auditor Project Proposal

1. Project Title

Network Auditor: Mobile Network Discovery, Vulnerability Assessment, and Security Intelligence Platform

2. Executive Summary

Network Auditor is a mobile-first cybersecurity platform designed for network administrators, security analysts, penetration testers, and IT support teams. The application provides a consolidated environment for network discovery, asset inventory, vulnerability assessment, port analysis, geolocation intelligence, and audit reporting.

The project addresses the growing need for quick, reliable, and field-ready network auditing on Android devices. The proposed system supports local and remote network inspection through scan orchestration, service fingerprinting, risk evaluation, and structured evidence collection. The platform will combine automated scanning with professional reporting to support incident response, compliance review, and infrastructure monitoring.

3. Background and Rationale

Modern enterprise networks include a wide mix of endpoints, servers, IoT equipment, wireless devices, and cloud services. These environments often present complex visibility gaps, inconsistent device tracking, unauthorized access points, and outdated software exposure. Traditional auditing workflows are frequently desktop-bound, time-consuming, and dependent on specialist toolchains.

Network Auditor addresses this gap by delivering a practical Android-based solution for continuous network visibility. The project builds on the current architecture of the Securenets Auditor application while expanding the core capabilities into a more formal, proposal-driven network audit system for institutional deployment.

4. Problem Statement

Organizations face persistent challenges in:

- Limited visibility of active devices on internal networks
- Delayed detection of unauthorized endpoints and rogue systems
- Inconsistent vulnerability assessment across distributed environments
- Fragmented evidence collection for audit and compliance review
- Restricted access to security analysis during field operations
- Weak integration between discovery, reporting, and intelligence collection

A mobile-responsive auditing platform is needed to simplify network evaluation, improve operational awareness, and support secure decision-making in the field and on the move.

5. Project Goal

Development of a comprehensive network auditing solution that supports discovery, assessment, reporting, monitoring, and intelligence gathering for network security operations.

6. Specific Objectives

The following specific objectives are written to comply with the stated constraint of no verbs, conjunctions, or adjectives:

- Discovery
- Inventory
- Exposure
- Fingerprint
- Scoring
- Intelligence
- Tracking
- Classification
- Reporting
- Mapping
- Vault
- History
- Workflow
- Correlation
- Storage
- Scheduling

7. Scope of the Project

7.1 In Scope

- Android-based network auditing interface
- Wi-Fi and subnet discovery
- Host scanning and port enumeration
- Service banner analysis
- Device manufacturer identification through MAC address lookup
- Risk scoring and vulnerability categorization
- Reporting in PDF or shareable formats
- Threat intelligence enrichment through public feeds and OSINT checks
- Secure storage of credentials and audit history
- Background monitoring for rogue device detection
- Dashboard summaries for network health and exposure overview

7.2 Out of Scope

- Full enterprise SIEM deployment
- Cloud-native firewall orchestration
- Deep packet inspection at full traffic volume
- Legal penetration testing authorization management
- Third-party hardware sensor deployment
- Complete enterprise identity management integration

8. Proposed Features and Functional Modules

8.1 Network Discovery Module

This module identifies the active subnet, collects reachable hosts, and compiles a device inventory using ICMP, ARP, and socket-based probing methods. It will present a map of discovered assets with IP, MAC, vendor, and status indicators.

8.2 Port Scanning and Service Identification

The system will identify open ports, capture service banners, and classify services by protocol and software signature. This feature supports reconnaissance for weak services, outdated systems, and exposure patterns.

8.3 Vulnerability Assessment Engine

Network Auditor will evaluate collected evidence against known security patterns. Risk scoring will be based on service exposure, application version exposure, protocol weakness, and intelligence correlation.

8.4 OSINT and Threat Intelligence Layer

The platform will integrate external lookups for domain reputation, malware indicators, breach status, geolocation, and abuse telemetry. This supports contextual awareness beyond internal scanning.

8.5 Reporting and Documentation Module

Users will generate professional audit reports containing scan summaries, findings, risk scores, evidence, and recommendations. Export options may include PDF reports for archiving and review.

8.6 Secure Storage and Credential Vault

Sensitive operational artifacts, API keys, credentials, and audit records will be stored with encryption and protected access controls. Biometric authentication may be used to reinforce privacy and controlled access.

8.7 Monitoring and Alerting Module

The system will support periodic scans for new devices, offline host detection, and suspicious changes in environment baseline. Alerts will help reduce response time for unauthorized activity.

9. Technical Approach

The proposed solution will follow a modular Android architecture with the following components:

- Kotlin-based Android application
- Jetpack Compose for UI implementation
- MVVM architecture for separation of concerns
- Repository pattern for data handling
- Room database for scan history and asset records
- Retrofit for external API integration
- Coroutines and Flow for asynchronous scanning workflows
- EncryptedSharedPreferences and secure local encryption for sensitive settings
- Support for Wi-Fi and network permission-based scanning

10. System Architecture Overview

The system architecture will include the following layers:

1. Presentation Layer

- Dashboard
- Scanner controls
- Report viewer
- Intelligence panel
- Settings and vault access

2. Application Layer

- Scan orchestration service
- Risk evaluation engine
- Report generator
- Threat feed manager
- Notification manager

3. Data Layer

- Local scan history database
- Cached assets and findings
- Secure credentials store
- External intelligence API cache

4. Network Layer

- Host discovery sockets
- ICMP and ARP probes
- Port scanning routines
- API calls for geolocation and reputation data

11. Development Methodology

The project will use an iterative development approach with the following phases:

- Requirement analysis and feature prioritization
- Core architecture setup and database schema design
- Discovery and scan module implementation
- Risk analysis and reporting module implementation
- Threat intelligence integration
- Security hardening and performance optimization
- User testing and deployment preparation

12. Project Deliverables

The project will deliver the following outputs:

- Complete Android application source code
- Network discovery and asset inventory workflow
- Vulnerability analysis engine

- PDF report generation module
- Secure vault components
- Monitoring and alerting mechanism
- User documentation and operator guide
- Deployment checklist and maintenance notes

13. Implementation Timeline

Phase 1: Planning and Design (Weeks 1-2)

- Requirements confirmation
- Architecture definition
- Data model specification
- User flow planning

Phase 2: Core Development (Weeks 3-6)

- Scanner integration
- Asset inventory functions
- Port analysis modules
- Initial reporting workflow

Phase 3: Intelligence and Security Layer (Weeks 7-9)

- OSINT integration
- Risk scoring implementation
- Encryption and secure storage
- Notification handling

Phase 4: Testing and Deployment (Weeks 10-12)

- Functional testing
- Security validation
- User acceptance testing
- Packaging and release preparation

14. Resource Requirements

Human Resources

- Project manager
- Android developer
- Backend/API integration engineer
- Security analyst consultant
- QA tester
- UI/UX designer

Technical Resources

- Android Studio environment

- Kotlin development tools
- Android device or emulator for testing
- Network lab environment for validation
- Secure storage and encryption configuration
- External intelligence API access

15. Budget Estimate

A preliminary budget estimate for the project is outlined below:

- Development personnel: \$18,000 - \$30,000
- Security testing and validation: \$4,000 - \$8,000
- API integration and intelligence services: \$2,000 - \$5,000
- Design and documentation: \$2,000 - \$4,000
- Contingency reserve: \$3,000 - \$6,000

Estimated total: \$29,000 - \$53,000

16. Risk Assessment and Mitigation

Potential Risks

- Permission limitations on Android device scanning
- Network congestion during large scans
- External API rate limits or downtime
- Inaccurate fingerprinting from encrypted services
- User trust concerns around monitoring tools

Mitigation Strategies

- Permission-aware feature gating and user guidance
- Adaptive scan timing and rate control
- Local caching and fallback methods for intelligence queries
- Service fingerprinting with multiple evidence sources
- Transparent privacy controls and audit logging

17. Expected Outcomes

The proposed project is expected to produce:

- Improved network visibility for administrators
- Faster discovery of suspicious devices and exposures
- Structured evidence collection for audit processes
- Better prioritization of security review tasks
- Stronger awareness of risk posture across internal networks
- A mobile platform suited for progressive network auditing workflows

18. Sustainability and Maintenance Plan

The solution will require periodic maintenance to preserve compatibility with evolving Android versions, network protocols, and API interfaces. Maintenance planning should include:

- Update cycle for scanning routines
- Intelligence source review and refresh policy
- Bug tracking and incident response process
- Security patch review for encryption and permissions
- Release schedule for feature expansion and usability enhancement

19. Conclusion

Network Auditor represents a practical, scalable, and security-focused proposal for modern network auditing on mobile devices. The system merges discovery, scoring, intelligence, and reporting into a unified workflow, enabling faster visibility into network risks and stronger operational decision-making. The project offers a realistic path toward a production-quality security auditing platform suitable for field operations, internal audits, and continuous infrastructure awareness.